

RIPHAH INTERNATIONAL UNIVERSITY, ISLAMABAD



Assignment MS MPI

Bachelors of Computer science – 7th semester

Subject: Parallel and Distributed Computing

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MS MPI Downloads

The following are current downloads for MS MPI:

Download Microsoft MPI v10.1.3 from Official Microsoft Download Center

Download MS MPI execution file and Sdk

Requirements:

Visual Studio (with C++ Distribution)

MS MPI , SDK

Steps for running a MS MPI Program on two nodes

We Create a New C++ project

New Project -> Windows Desktop Application

The name of the project and files should be the same for two machines.

Write a Basic program of MS MPI which displays total processes, the current running process and processor name

Here is the code:

Insert code in c++ file:

```
#include <iostream>
#include <mpi.h>
#include <vector>

int main(int argc, char** argv) {
    MPI_Init(&argc, &argv);

    int world_size;
    MPI_Comm_size(MPI_COMM_WORLD, &world_size);

    int my_rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &my_rank);

    char processor_name[MPI_MAX_PROCESSOR_NAME];
    int name_len;
    MPI_Get_processor_name(processor_name, &name_len);

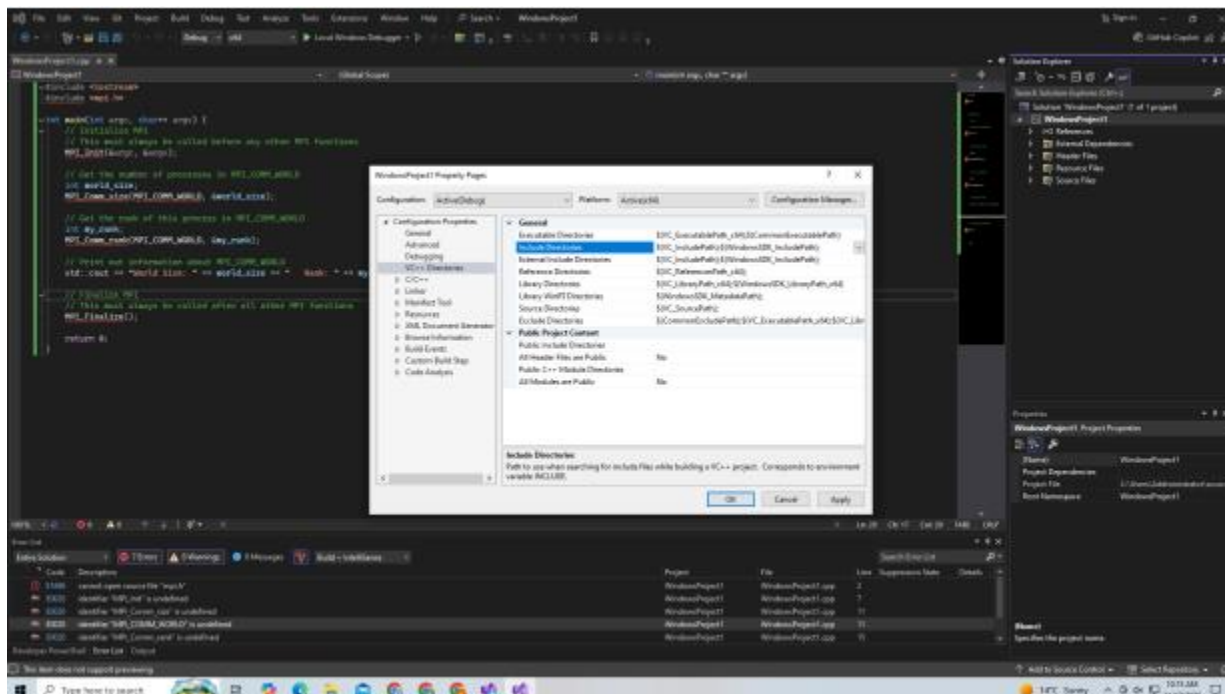
    std::cout << "Process " << my_rank << " of " << world_size
                << " is running on machine: " << processor_name << std::endl;

    MPI_Finalize();

    return 0;}

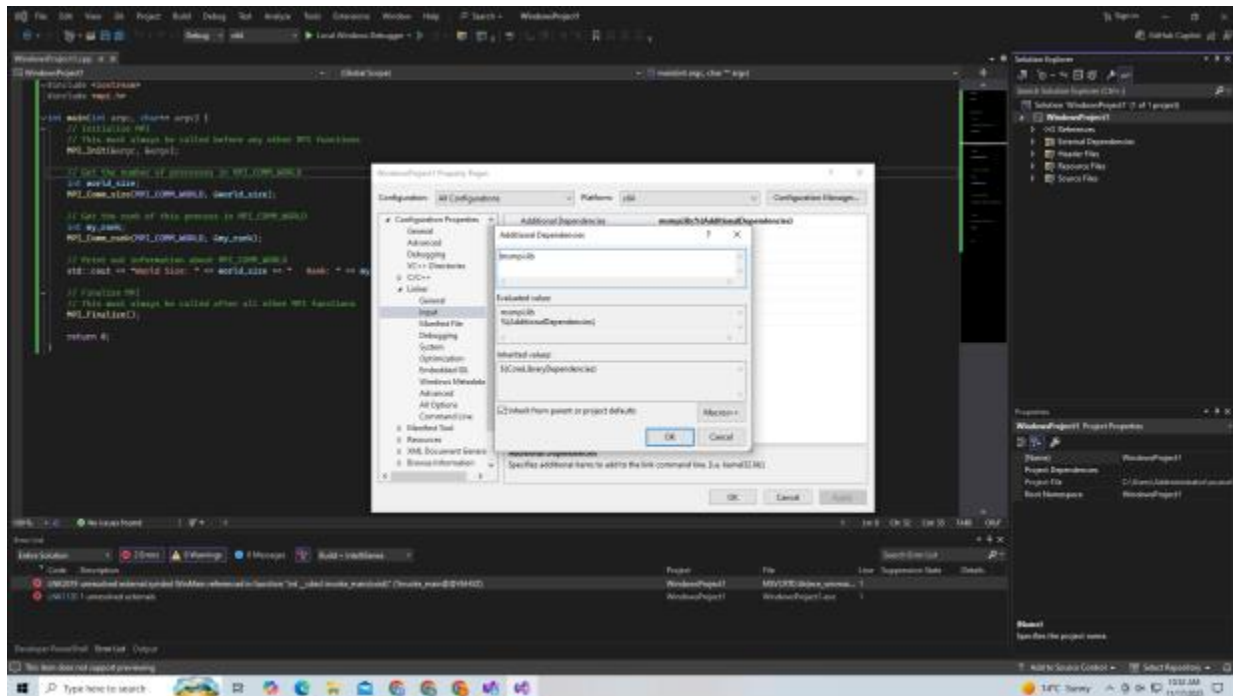
```

Add paths of MS MPI in project properties as given below:



- In Additional directories insert

msmpi.lib



In linker systems, choose subsystem and from dropdown select subsystem console:

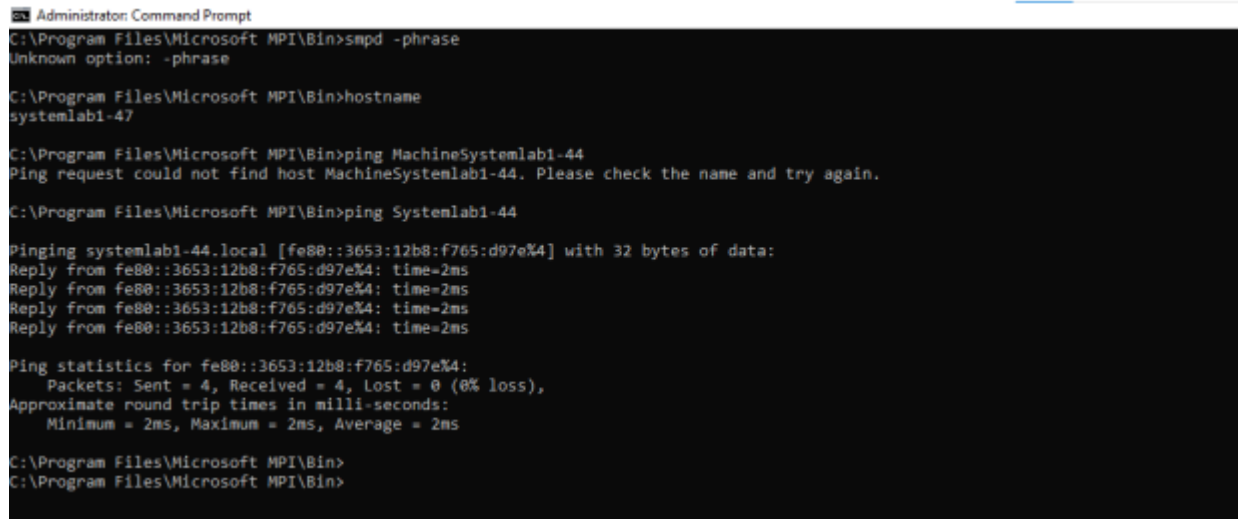
Commands:

netsh advfirewall set allprofiles state off

Note: Smpd is a cluster process manager

For running on multiple machines, two machines should be logged in the same accounts or should do smpd settings.

Use ping command to check connection:



```
Administrator: Command Prompt
C:\Program Files\Microsoft MPI\Bin>smpd -phrase
Unknown option: -phrase

C:\Program Files\Microsoft MPI\Bin>hostname
systemlab1-47

C:\Program Files\Microsoft MPI\Bin>ping MachineSystemlab1-44
Ping request could not find host MachineSystemlab1-44. Please check the name and try again.

C:\Program Files\Microsoft MPI\Bin>ping Systemlab1-44

Pinging systemlab1-44.local [fe80::3653:12b8:f765:d97e%4] with 32 bytes of data:
Reply from fe80::3653:12b8:f765:d97e%4: time=2ms
Reply from fe80::3653:12b8:f765:d97e%4: time=2ms
Reply from fe80::3653:12b8:f765:d97e%4: time=2ms
Reply from fe80::3653:12b8:f765:d97e%4: time=2ms

Ping statistics for fe80::3653:12b8:f765:d97e%4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Program Files\Microsoft MPI\Bin>
C:\Program Files\Microsoft MPI\Bin>
```

Create a machine file (txt) to run it on multiple machines, Add machines host names e.g systemlab2-21

Command for execution using different machines

```
Mpiexec -machinefile machines.txt -np 4 MSMPI.exe
```

